

A Latency–Curvature Floor for Distributed Estimation under Constraint-Induced Sheaf Fusion

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TABLE I
SWITCH-OFF LADDER (TIER-1 ROWS; EXECUTED VALUES).

arm	prediction	measured
generic slope	2	1.999
coefficient ratio ($m=2$)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
$\eta = 0$	0	$< 10^{-17}$
$\xi = 0$	0	0.0

Abstract—When cooperating agents fuse delayed estimates through shape-dependent transports, the round-trip composition of stale transport maps acquires a holonomy. We prove that, to leading order in the staleness τ , the error-transport holonomy amplitude obeys $\|\text{Log Hol}\| = \tau^2 \| [C_i, C_j] \| + O(\tau^3)$, with C_j the shape-conjugated error generators, vanishing identically under each of the theorem’s hypotheses’ negations ($\eta = 0$, $\xi = 0$, coincident shapes). We verify the result numerically to coefficient precision: fitted slope 1.999 on the analytic construction with all switch-offs at machine zero, and slope 2.000 with the $m=2$ coefficient ratio 1.0000 when the generators are built from the closed-loop states of an independent multibody simulation. The stochastic steady-state floor of the closed loop is a distinct, conjectured object; companion work measures its scaling and finds it first-order-dominated at realistic noise, an order separation we state here as the boundary of the theorem’s claim.

I. INTRODUCTION

[**Author section:** motivation from the companion theory — cable-coupled multi-agent estimation, the constraint channel, delayed fusion through shape-conjugated transports, and why the round trip is the natural object. Source: estimation_sheaf.tex §1.]

II. SETTING AND MAIN RESULT

[**Author section:** Theorem 7.2 stated verbatim with its tag — “(leading order, deterministic; sharp constants and the stochastic steady-state floor: [Conjectural])” — the hedge is load-bearing; plus the supporting BCH lemma (Lem. 7.1). Source: estimation_sheaf.tex §7.]

III. THE MEASURED OBJECT

The $m=2$ round trip composes one edge’s error transport both ways: $\text{Hol} = e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$, with $C_j = \text{Ad}_{m(s_j)} \text{ad}_\xi \text{Ad}_{m(s_j)}^{-1}$. Lemma 7.1 gives $\text{Log Hol} = \tau^2 [C_i, C_j] + O(\tau^3)$.

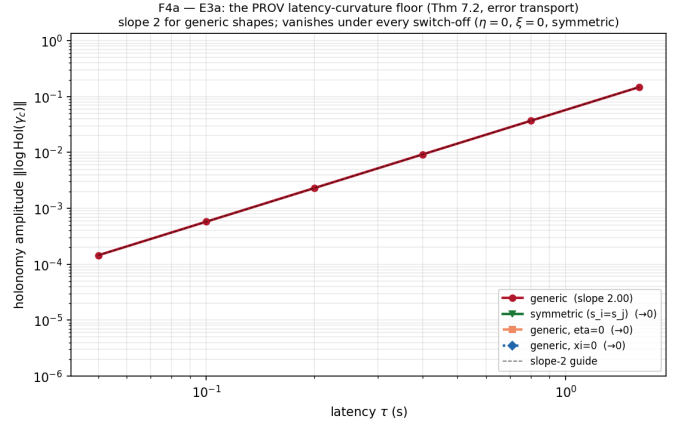


Fig. 1. Measured $\|\text{Log Hol}\|$ vs. τ (analytic shapes). Generic pair: fitted slope 1.999 over $\tau \in [0.05, 1.6]$ s; the coefficient ratio $\|\text{Log Hol}\|/\tau^2 \| [C_i, C_j] \| = 1.0000$ at the smallest τ . Switch-off arms ($s_i=s_j$; $\eta=0$; $\xi=0$) sit at machine zero ($\xi=0$ literally 0.0) — the effect dies exactly when the theorem requires it to.

a) *Independent-plant check.*: Building the same generators from the closed-loop shape and twist *estimates* of a physics-complete multibody simulation (Drake; SAP contact; 12 m distance-constraint cables; five-vessel pentagon tow) gives slope 2.000, coefficient ratio 1.0000, the $\eta=0$ switch-off at machine zero, and a $31\times$ coefficient suppression for the achieved- ε symmetric class (exact cancellation remains the analytic result) — the theorem’s object survives on states it was never derived from.

IV. WHAT THE THEOREM DOES NOT CLAIM

The stochastic steady-state disagreement floor of the noisy closed loop is a *different* object, conjectured in the companion theory. Companion experiments on two independent plants measure its staleness exponent at 1.08–1.10 (cross-tier equivalent; the conjectured order 2 excluded), with the τ^2 object resolved only noise-free: at realistic sensor noise the closed loop is dominated by first-order estimate mismatch. We state this boundary here so the theorem is cited for what it proves.

V. REPRODUCIBILITY

Every number above traces to a seeded, versioned driver; campaign_replay.py in the companion repository re-executes per-run probes that must match committed records exactly.